

SECJ3483 Web Technology

Group Project Brief

Academic Session 2025/2026 • Bachelor of Software Engineering with Honours

Group project assessment focusing on full-stack integration, RESTful API development, and secure web application deployment.

Course Code	SECJ3483
Course Name	Web Technology
Group Size	4 students per group
Project Weight	20% (Proposal 5% + Progress 5% + Demo 5% + Peer 5%)

1. Project Purpose and Alignment

This group project requires each team to design, develop, and present a full-stack web application that reflects the core topics covered in SECJ3483 Web Technology. The revised scope is intentionally aligned to the CI, particularly frontend development with JavaScript and Vue.js, asynchronous communication using AJAX and API calls, backend development using PHP Slim, MySQL database integration, full CRUD implementation, token-based authentication using JWT, and secure deployment practice.

The project is intended to measure students' ability to integrate frontend and backend components into a complete web application while demonstrating sound design, teamwork, and professional practice.

2. Topic Focus

- Front-end fundamentals: HTML, CSS, responsive layout, JavaScript DOM manipulation, and event handling.
- jQuery and AJAX: dynamic DOM interaction and asynchronous communication with APIs.
- Vue.js: directives, components, routing, state handling, forms, validation, and SPA workflow.
- RESTful backend using PHP Slim: route design, middleware, JSON request/response handling, and CRUD operations.
- Database integration using MySQL and PDO with at least three related entities.
- Security and integration: JWT-based authentication, token validation, protected routes, input validation, and basic defense against SQL injection and other common web threats.
- Deployment: hosting the integrated full-stack application on an accessible server environment.

3. Required Project Scope

Each group must develop one integrated application that satisfies all of the following minimum requirements:

- A clear problem domain and identified target users.
- Single Page Application (SPA) behaviour using Vue.js.
- Frontend pages that consume backend services asynchronously.
- A complete REST API with GET, POST, PUT, and DELETE operations.
- At least three related database tables or collections with realistic sample data.
- At least one authentication flow using JWT for login and protected access.
- Input validation on both frontend and backend.
- Basic error handling with suitable HTTP status codes.
- Deployment to a live or test server for demonstration.

4. Suggested Project Themes

Teams may propose their own case, provided it remains feasible within the semester and demonstrates the required CI topics.

5. Assessment Structure

The group project contributes 20% to the course grade and is divided into four components, each carrying 5%.

No.	Component	Assessment Form	Weight
1	Project Proposal	Group written submission	5%
2	Project Progress	Group progress review and checkpoint demo	5%
3	Project Demo	Final live demo and presentation	5%
4	Peer Evaluation	Individual peer rating within group	5%

6. Proposal Rubric (5%)

The proposal should establish the feasibility, structure, and direction of the project. It must be submitted as a concise but complete planning document.

Criterion	Expected Evidence	Marks	Weight
Problem statement and user focus	Problem is relevant, clearly defined, and matched to specific target users.	1.0	20%
Functional scope and modules	Core modules are realistic, coherent, and sufficient for a full-stack semester project.	1.0	20%
Technical design	Architecture diagram, API concept, database schema, and stack selection are suitable.	1.5	30%
Interface planning	Wireframes or mockups show main SPA pages and navigation flow.	0.75	15%
Team planning	Roles, responsibilities, and timeline are clear and balanced.	0.75	15%

7. Progress Rubric (5%)

The progress checkpoint focuses on front-end development first. At this stage, teams are evaluated on the completion, usability, and organisation of the Vue.js single page application, including interface structure, routing, forms, validation, and use of asynchronous communication on the front end. Back-end completeness, database integration, and JWT security are not the main emphasis for this checkpoint.

Criterion	Expected Evidence	Marks	Weight
SPA structure and navigation	Main pages or views are available and organised as a Single Page Application using Vue.js routing or equivalent navigation flow.	1.25	25%
Interface implementation	Frontend screens are clearly designed, usable, and reflect the proposed modules or user tasks.	1.25	25%
Forms and validation	Key input forms are working with suitable client-side validation and clear user feedback.	1.0	20%
Frontend async interaction	Frontend demonstrates at least one meaningful asynchronous interaction such as fetching, displaying, filtering, or submitting data using AJAX, Fetch, Axios, or similar.	1.0	20%
Version control and teamwork	The repository is active and contributions are visible across members.	0.5	10%

8. Demo Rubric (5%)

The final demo assesses the quality of the completed application and the group's ability to explain, justify, and demonstrate its implementation.

Criterion	Expected Evidence	Marks	Weight
SPA frontend quality	Application flow is coherent, responsive, and implemented as a working SPA.	0.75	15%
CRUD completeness	Core data operations using GET, POST, PUT, and DELETE work correctly.	1.0	20%
REST API implementation	Routes, request handling, JSON responses, and HTTP status codes are correctly managed.	0.75	15%
Database design and use	Database has meaningful related entities and supports the implemented features.	0.5	10%
JWT and protected access	Authentication flow and token-protected functionality are demonstrated clearly.	0.75	15%
Validation and security practice	Input validation, error handling, and awareness of security threats are evident.	0.5	10%
Presentation and technical explanation	Group explains architecture, design choices, and implementation confidently.	0.75	15%

9. Peer Evaluation Rubric (5%)

Peer evaluation is completed individually and confidentially. Each student evaluates the contribution of their teammates.

Criterion	3 - Excellent	2 - Satisfactory	1 - Needs Improvement	Max
Task completion	Always completed assigned tasks on time.	Completed most tasks with acceptable consistency.	Frequently delayed or missed assigned tasks.	3
Contribution quality	Produced high-quality and useful work for the group.	Work was acceptable but uneven in quality.	Contribution was minimal or weak.	3
Communication	Communicated clearly and proactively throughout the project.	Communicated when needed or when prompted.	Rarely communicated or caused coordination problems.	3
Reliability and teamwork	Fully dependable and cooperative in team activities.	Generally dependable with minor issues.	Often unreliable or difficult to work with.	3

Peer evaluation score conversion: average score obtained $\div 12 \times 5$.

10. Deliverables

- Proposal document in PDF or Word format.
- Source code repository link with visible group contribution history.
- Presentation slides for final demo.
- Live URL or hosted test environment, where applicable.
- Database schema, export, or seed file.
- README with setup instructions.
- Completed peer evaluation form.

11. Important Date

Date	Output
Weeks 5	Group formation, topic selection, early requirements, role distribution
3 May 2026	Proposal submission
Week 11 (25 – 31 May 2026)	Front-end Progress Evaluation (during class session)
Week 15 (22 – 28 June 2026)	Demo and presentation (during class session)
30 June 2026	Peer evaluation

12. Academic Integrity and Expectations

- All submitted work must be original and explainable by the group members.

- Use of frameworks and libraries is permitted, but the team must understand the code used.
- AI-assisted coding may be used only as a learning support and not as a substitute for understanding.
- Every group member must participate actively in analysis, development, and presentation.
- The lecturer may question any member during the demo to verify individual contribution and understanding.